Exercise 1: Let C1 and C2 be two consistent cuts. Show that the intersection of C1 and C2 is a consistent cut.

def A cut C is consistent iff: $e \longrightarrow e' \wedge e' \in C \Rightarrow e \in C$

Let $e' \in C = C_1 \cap C_2$ and e t.c. $e \longrightarrow e'$. By definition, since $e' \in C_1 \cap C_2$ we also have $e' \in C_1$ and $e' \in C_2$.

Howevevre, since $e \longrightarrow e'$ and C_1 is a consistent cut, e also HUST be in C_1.
Analog reasoning for C_2. Because $e \longrightarrow e'$ and C_2 is consistent, $e \in C_2$.
At this point, $e \in C_1 \wedge e \in C_2 \Rightarrow e \in C = C_1 \cap C_2$

So, $\forall e, e' \quad e' \in C = C_1 \cap C_2 \wedge e \longrightarrow e' \Rightarrow e \in C_1 \cap C_2 \Rightarrow C_1 \cap C_2$ CONSISTENT

Exercise 2: Let C1 and C2 be two consistent cuts. Show that the union of C1 and C2 is a consistent cut.

def A cut C is CONSISTENT iff: $e' \in C \wedge e \longrightarrow e' \Rightarrow e \in C$

Let $e' \in C = C_1 \cup C_2$ and e t.c. $e \longrightarrow e'$.

By definition of "consistent cut", since $e' \in C_1$ and $e \longrightarrow e'$, we have that also $e \in C_1$.
And $e \in C_1 \Rightarrow e \in C = C_1 \cup C_2$. (Same for C_2)

So, $\forall e' \in C = C_1 \cup C_2 \wedge e \longrightarrow e' \Rightarrow e \in C = C_1 \cup C_2 \Rightarrow C_1 \cup C_2$ CONSISTENT

Exercise 3: Show that every consistent global state can be reached by some consistent run.

Let E be the set of all events in the system. Exploiting the "happened before" relationship ($\longrightarrow$) we can create a sorting algorithm $sort(\cdot)$ such that:

- if $e \longrightarrow e'$ then e is placed before e'

Then we can apply this sorting algorithm to C and $E \setminus C$, in sequence, to create a run R:

$R = sort(C)$ followed by $sort(E \setminus C)$

At this point, R is also consistent because $\forall e, e' \in E$ t.c. $e \longrightarrow e'$:

- if $e \in C$ and $e' \in C$ then e will be placed before e' (by definition of $sort(\cdot)$)
- if $e \in C$ and $e' \in E \setminus C$ then e will be placed before e', since e will be in $sort(C)$ and e' will be in $sort(E \setminus C)$
- if $e \in E \setminus C$ and $e' \in C$ then C would not be consistent. So, this case is not possible
- if $e \in E \setminus C$ and $e' \in E \setminus C$ then e will be placed before e' (by definition of $sort(\cdot)$)

Exercise 4: Let C1 and C2 be two consistent cuts. If C1 is a subset of C2, then C2 is reachable from C1. (There exists a consistent run that reaches C1 and then reaches C2.)

Let E be the set of all events in the system. Exploiting the "happened before" relationship ($\longrightarrow$) we can create a sorting algorithm $sort(\cdot)$ such that:

- if $e \longrightarrow e'$ then e is placed before e'

Since $C_1 \subseteq C_2$ we can divide E into three disjoint sets: C_1, $C_2 \setminus C_1$ and $E \setminus C_2$. Then we can apply this sorting algorithm to C_1, $C_2 \setminus C_1$ and $E \setminus C_2$, in sequence, to create a run R:

$$R = sort(C_1) \text{ followed by } sort(C_2 \setminus C_1) \text{ followed by } sort(E \setminus C_2)$$

At this point, R is also consistent because $\forall e, e' \in E$ t.c. $e \longrightarrow e'$:

- if e and e' are in the same set (both in C_1, both in $C_2 \setminus C_1$ or both in $E \setminus C_2$) then e will be placed before e' (by definition of $sort(\cdot)$)
- if e and e' are in different sets "left to right" ($e \in C_1 \wedge e' \in C_2 \setminus C_1 \cup E \setminus C_1$ or $e \in C_2 \setminus C_1 \wedge e' \in E \setminus C_2$) then e will be placed before e' by construction of R ($sort(C_1)$ appears before $sort(C_2 \setminus C_1)$ that appears before $sort(E \setminus C_2)$)
- if e and e' are in different sets "right to left", C_1 or C_2 would not be consistent. So this case is not possible.

Exercise 5: Label with proper logical clock all the events of the distributed computation in image vector.pdf. (You can consider events that receive a message and immediately send it as single events.)

